PROOF OF CONCEPT REPORT

Electronic Medicine Approval System with eCTD Technical Validation

تقرير إثبات المفهوم: نظام إلكتروني للموافقة على الأدوية مع التحقق الفني بصيغة eCTD

Document reference	PPR-POC-001
Version / status	0.2 · Draft for review
Date of issue	21 October 2026
Prepared for	Head of Department, Pharmacy
Prepared by	PPR project team
Classification	Internal

Revision history

Version	Date	Description	Author
0.1	14 Oct 2026	First issue of the proof of concept report	PPR project team
0.2	21 Oct 2026	Updated for specialist review teams, GCC Module 1 validation, the official ICH DTD, dossier tools and security additions. New section 7: workflow training guide. All screens recaptured.	PPR project team

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1 Executive summary

This report presents a working proof of concept (POC) for an electronic medicine approval system, named PPR. Pharmaceutical companies submit marketing authorisation dossiers in the international eCTD format. The system checks every package automatically, as the FDA and EMA do, before any scientific review starts.

The POC covers the complete process: submission, technical validation, assessment by a team of specialist reviewers (one per CTD module, led by a lead reviewer), requests for information, the Head of Department's decision and a bilingual approval certificate. It operates in English and Arabic and is ready to deploy on PythonAnywhere with a MySQL database.

Version 0.2 adds validation of the GCC regional Module 1 (v1.5) against the official ICH eCTD DTD, automatic assignment of review teams from reviewer specialties, reviewer tools (cumulative dossier view, notes in the PDF viewer, full-text search and a dossier assistant), stronger sign-in security, and an in-app **workflow training guide** that explains every workflow step by step for each role (section 7).

3 user roles	**10** application states	**29** validation rules in 3 layers	**2** languages (English, Arabic)
12 workflows in the training guide	**149** automated tests passed		

2 Background and objectives

2.1 Background

The FDA and the EMA receive drug applications as eCTD (electronic Common Technical Document) packages. Each submission is a folder called a *sequence* (0000, 0001, …) containing an XML backbone, `index.xml`, that lists every document in the five CTD modules together with its checksum.

Before the scientific content is assessed, each package must pass eCTD validation: an automated, rules-based technical check that the files are formatted correctly, are not secured, and are structured as required. The check has three layers: XML schema validation, business rule checks and file compliance inspection.

2.2 Objectives

- Receive medicine approval applications electronically in eCTD format.
- Apply the three validation layers automatically and give the applicant a clear report.
- Support the scientific review, requests for information and the final decision in one system.
- Keep a complete audit trail of every decision and status change.
- Provide an Arabic and English interface.

3 Scope of the proof of concept

3.1 In scope

- Company, reviewer, lead reviewer and Head of Department roles
- Application creation and eCTD zip upload
- Three-layer technical validation, including GCC Module 1, with reports
- Specialist review teams assigned from reviewer specialties
- Cumulative dossier view, PDF viewer with review notes, full-text search
- Module reviews, information requests and response sequences
- Decision, approval number and bilingual certificate
- Audit trail, in-app and email notifications
- Dossier assistant, read-only API, optional single sign-on
- In-app workflow training guide (English and Arabic)

3.2 Out of scope

- Fees and online payment
- Company self-registration and verification
- Write integration with other government systems (a read-only API exists)
- Two-factor login
- Post-approval variations and renewals linked to an approval

4 Business workflow

Every status change passes through a single state machine. It checks that the user's role is allowed to make the change, records it in the audit trail and notifies the people involved. Figure 1 shows the states and who triggers each transition. Once a dossier passes validation, the system assembles the review team from the reviewer specialties set by the Head of Department; the Head of Department assigns it by hand only when a required module has no qualified reviewer.

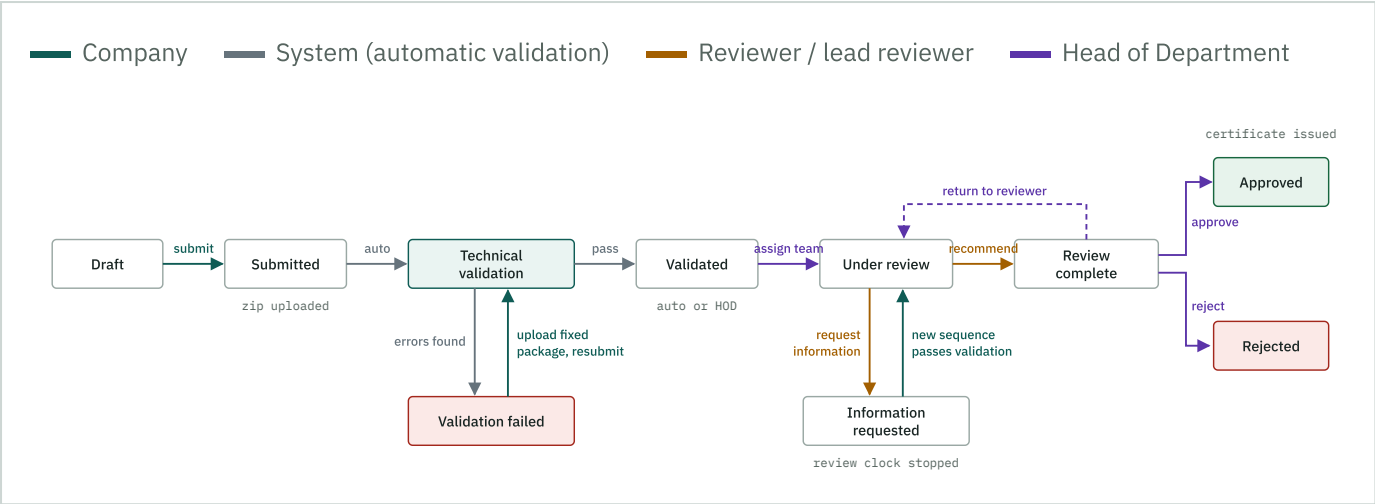


Figure 1: Application states and transitions. A package cannot reach a reviewer until it passes validation; answers to reviewer questions are new sequences and are validated in the same way. Questions and the recommendation come from the lead reviewer; module assessments from each specialist.

Role	Responsibilities in the system
Company applicant	Creates applications, uploads eCTD sequences, submits, reads validation reports, answers information requests and prints the certificate.
Reviewer	Specialised in CTD modules (M1–M5) set by the Head of Department. Reads the whole dossier of the applications on their team and records an assessment for the modules assigned to them. Keeps internal notes in the PDF viewer.
Lead reviewer	One member of the review team. Sends questions to the applicant (clock stop) and, once every module is assessed, the team's recommendation to the Head of Department.
Head of Department	Sets reviewer specialties and the default lead, monitors workload and status, adjusts review teams, approves or rejects, or returns the file to the lead reviewer.
System administrator	Manages user accounts, companies and roles, API keys and sign-in lockouts.

Table 1: User roles.

5 eCTD technical validation

Validation runs when the company submits. The package is unpacked safely, the XML is read once, and 29 rules run in three layers in a fixed order. Any error fails the sequence; warnings are reported but do not block it. `index.xml` is checked against the official ICH eCTD 3.2 DTD, and the regional Module 1 against the GCC Module 1 Specifications v1.5 with the envelope addressed to Oman (OM-

MOH). In production, validation runs in a background worker so large dossiers do not hold up the page.

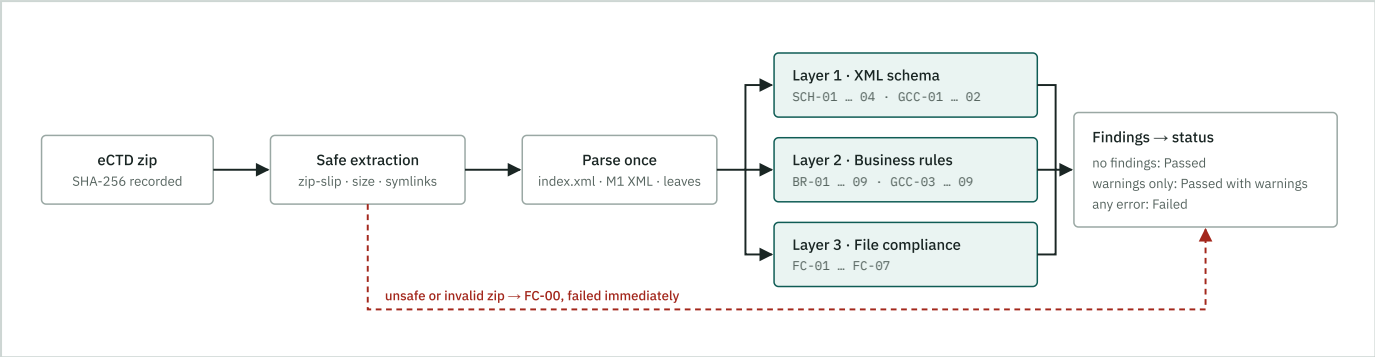


Figure 2: Validation pipeline. Failed packages are not kept unpacked; passed sequences become available in the dossier viewer.

Layer	Rule	Requirement	Severity
1. XML schema validation	SCH-01	The sequence contains `index.xml`	Error
	SCH-02	`index.xml` is well-formed XML	Error
	SCH-03	`index.xml` is valid against the ICH eCTD 3.2 DTD	Error
	SCH-04	Regional Module 1 XML (`m1/gc/gc-regional.xml`) exists and is well-formed	Error
	GCC-01	Its root is the GCC backbone in the GCC namespace	Error
	GCC-02	It is valid against the GCC regional DTD	Error
2. Business rule checks	BR-01	One sequence folder with the expected 4-digit number	Error
	BR-02	`index-md5.txt` matches the MD5 of `index.xml`	Error
	BR-03	Every document reference points to a file in the sequence	Error
	BR-04	Every document's MD5 checksum matches the declared value	Error
	BR-05	No files that the XML does not reference	Warning
	BR-06	Modules required for the application type contain documents	Error
	BR-07	Replace, append and delete operations refer to earlier accepted sequences	Error
	BR-08	ICH naming: lowercase letters, digits and hyphens; names up to 64 and paths up to 180 characters	Error
	BR-09	`util/dtd` contains the eCTD DTD	Error
	GCC-03	An envelope for the receiving country (or common) names its agency	Error
	GCC-04	Envelope sequence and related sequences match accepted sequences	Error
	GCC-05	Submission type fits the application type	Error

Layer	Rule	Requirement	Severity
	GCC-06	Submission unit fits the sequence (initial for 0000)	Error / warning
	GCC-07	Required Module 1 documents, including Arabic and English leaflets	Error
	GCC-08	`index.xml` references the regional XML with operation new	Error
	GCC-09	The GCC DTD files are in `util/dtd/`	Error
3. File compliance inspection	FC-00	The package is a valid, safe ZIP archive	Error
	FC-01	Accepted file formats only (.docx in Module 1 only)	Error
	FC-02	PDF files are not encrypted or password-protected	Error
	FC-03	PDF version is 1.4 to 1.7	Error
	FC-04	PDF files contain no JavaScript or launch actions	Error
	FC-05	Fonts in PDF files are embedded	Warning
	FC-06	No file exceeds the size limit (400 MB)	Error
	FC-07	No empty files	Error

Table 2: Validation rules by layer. FC-00 is the package safety check that runs before the 29 rules.

6 System screens

The following screens were captured on 21 October 2026 from the running prototype using demonstration data. Each image opens at full size.

6.1 Company applicant

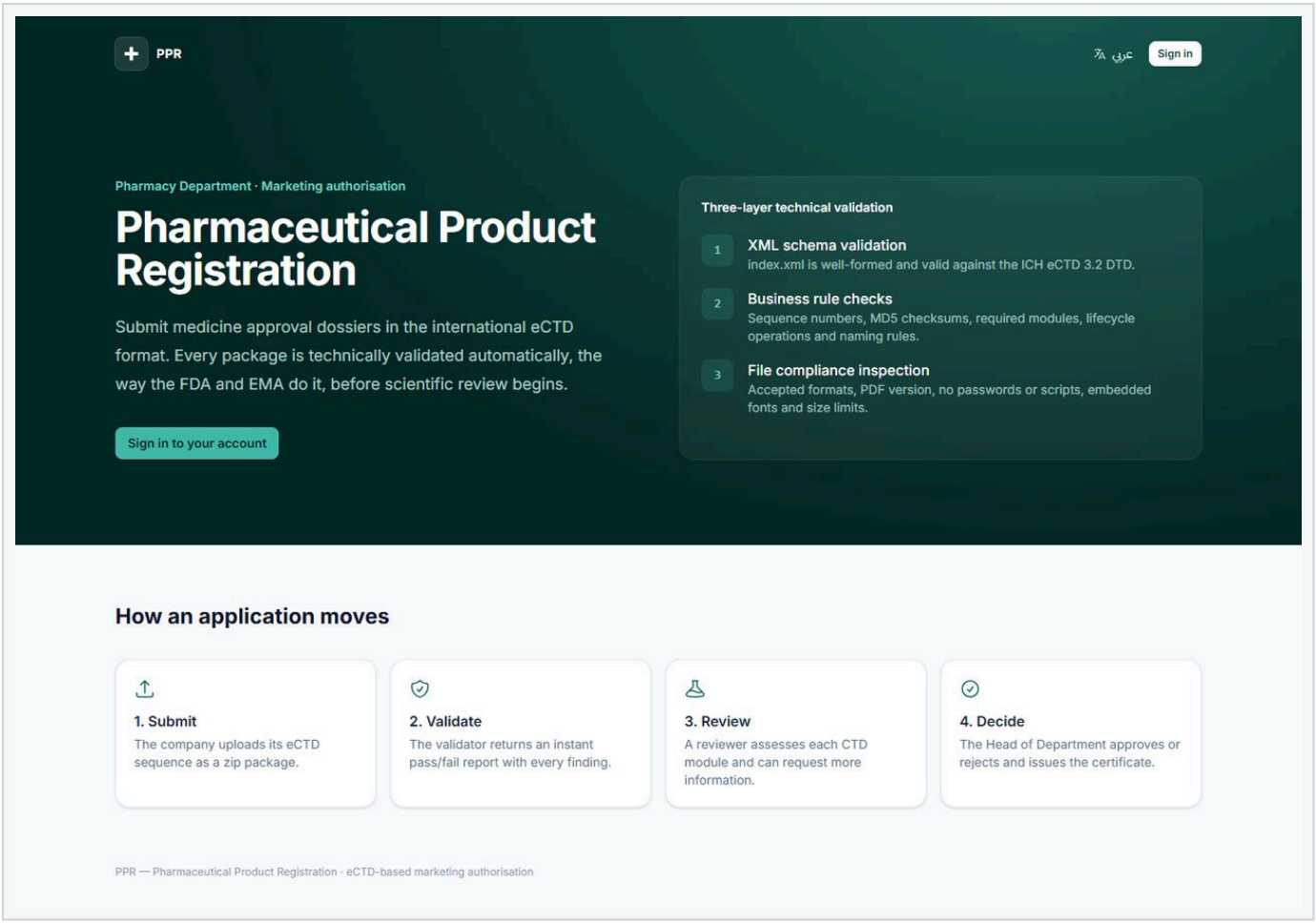


Figure 3: Public landing page describing the process and the three validation layers.

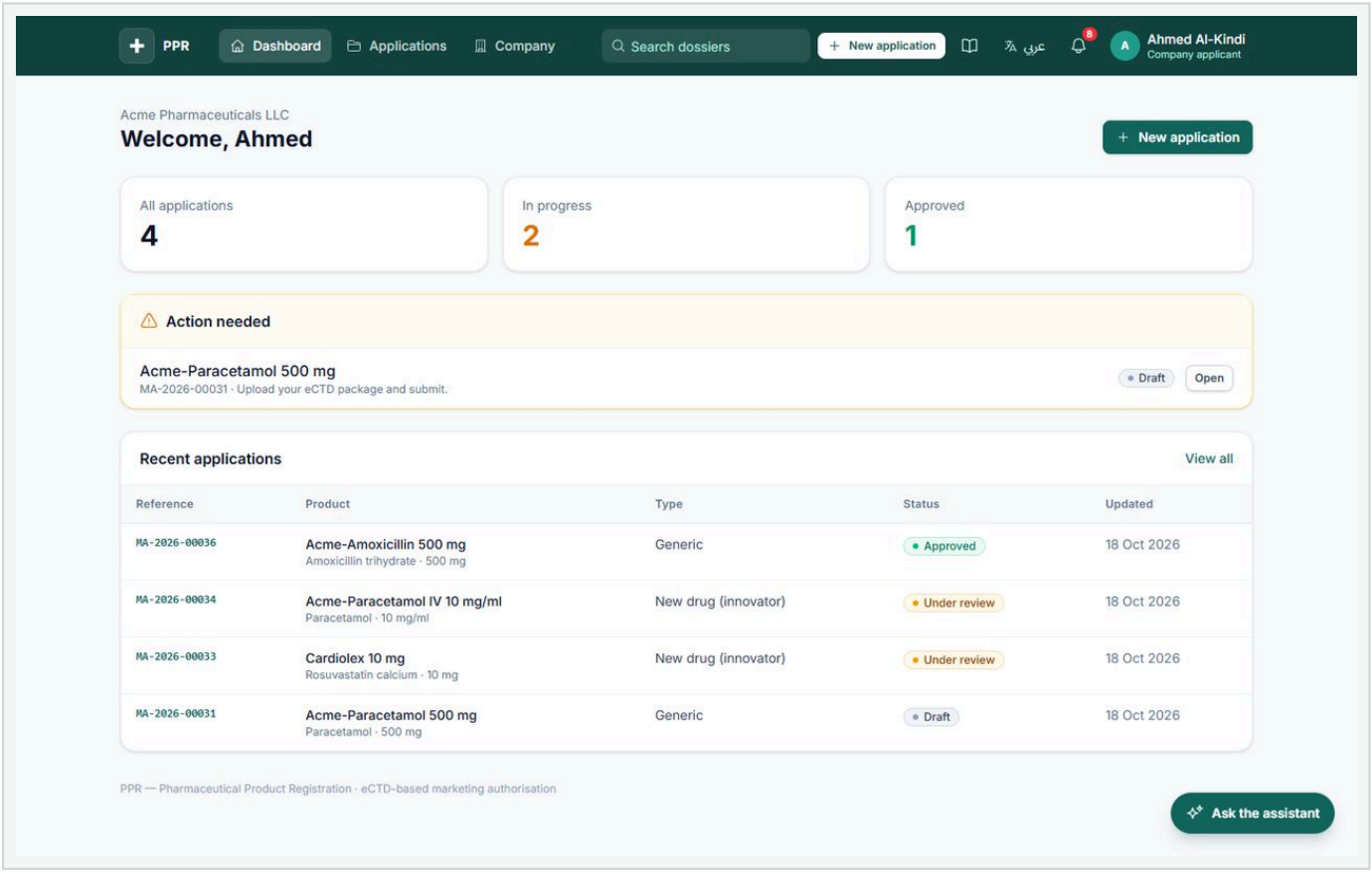


Figure 4: Company dashboard with totals, items needing action and recent applications.

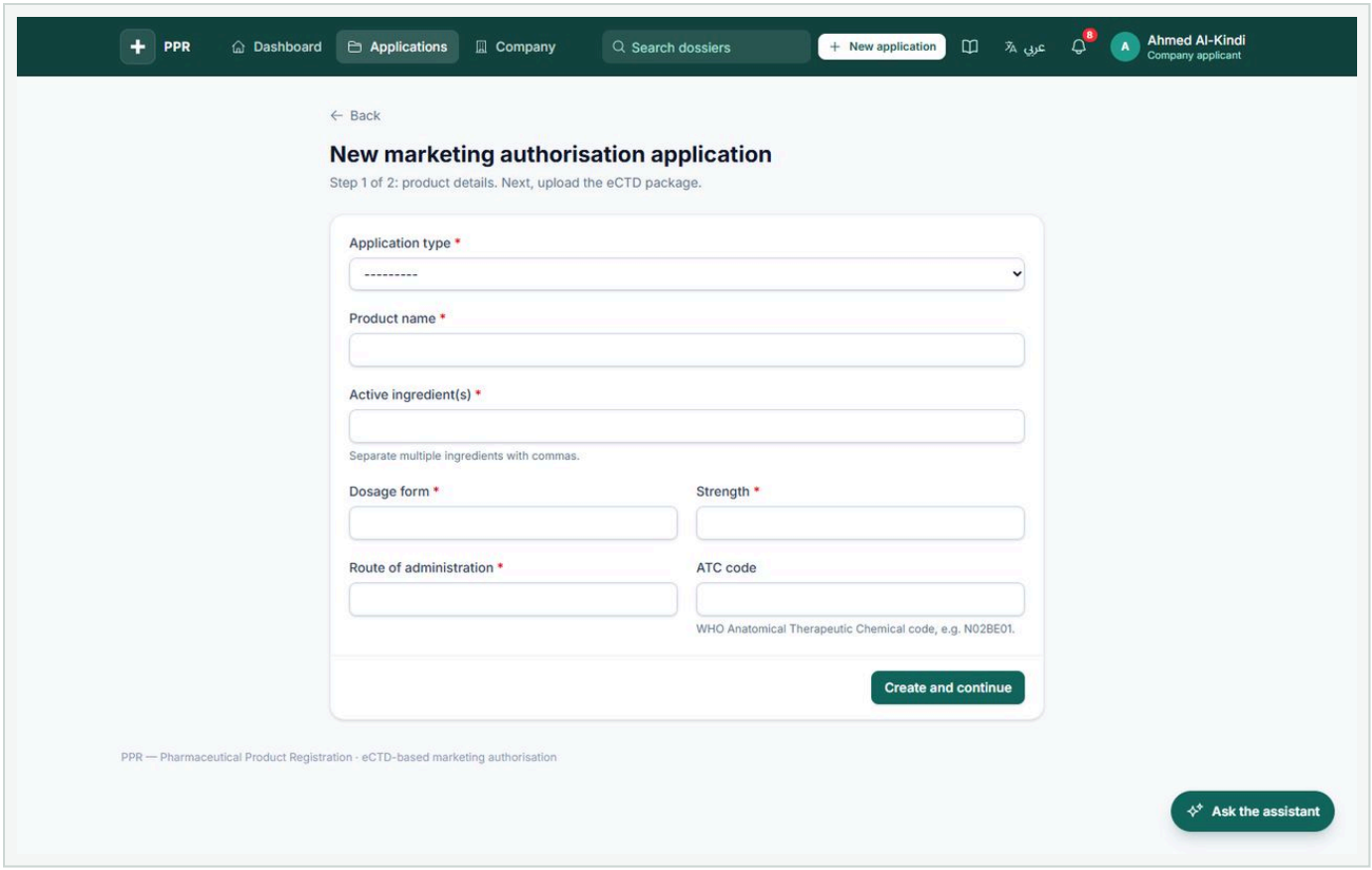


Figure 5: New application form for product details.

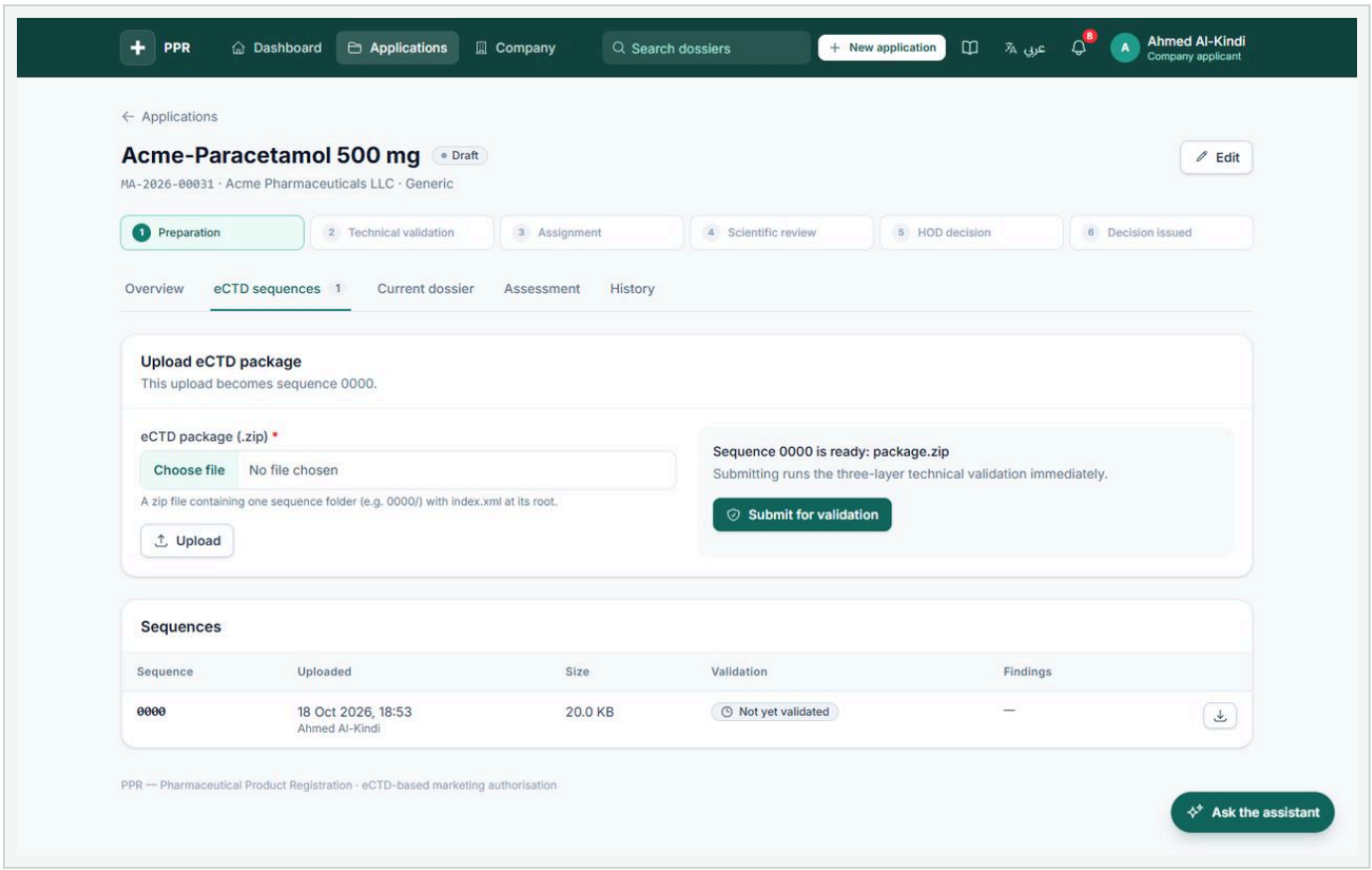


Figure 6: eCTD package upload with the package requirements. Submission starts validation immediately.

6.2 Technical validation

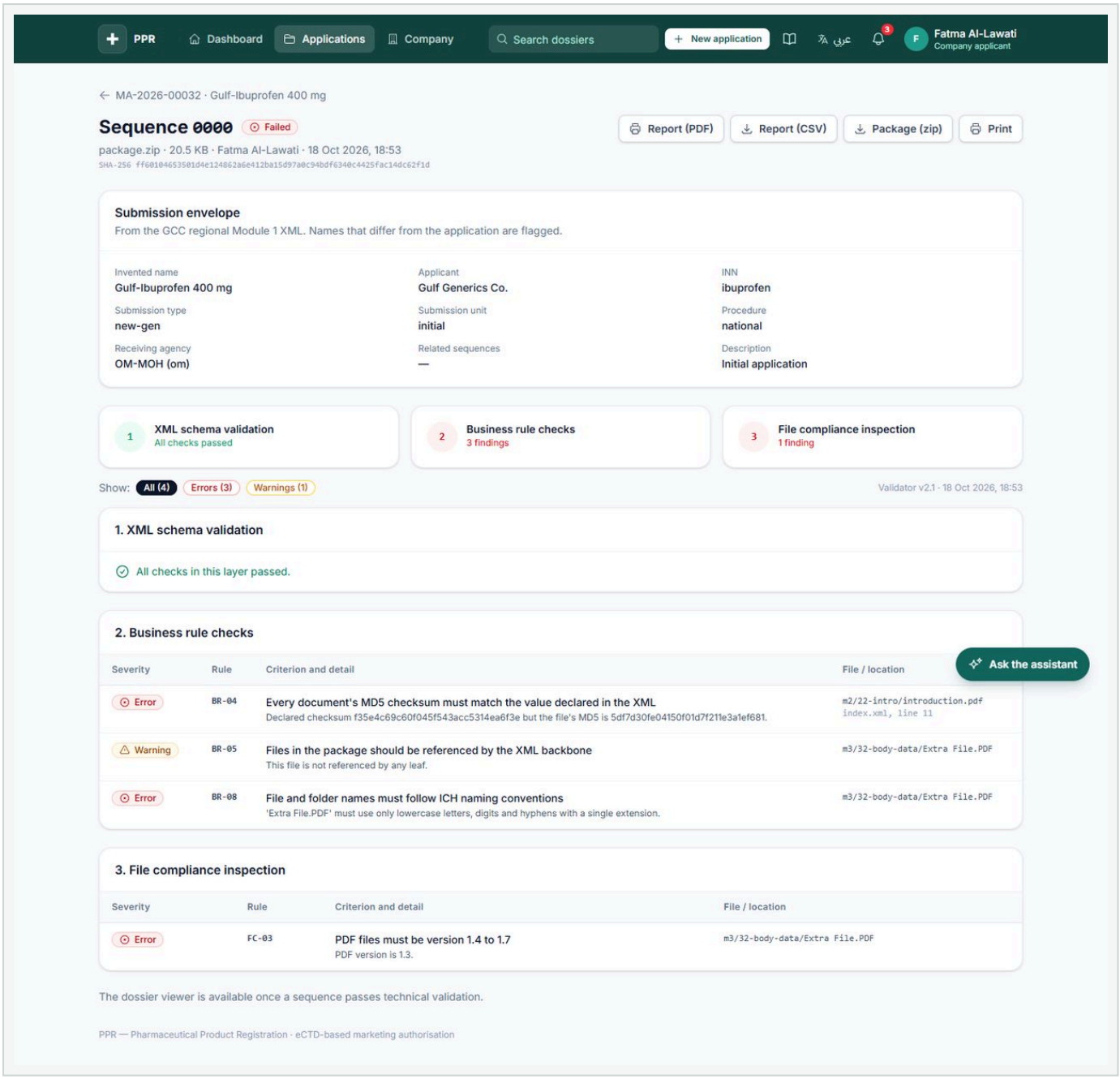


Figure 7: Failed validation report listing each finding with its rule, file and line, exportable as CSV or a printable report.

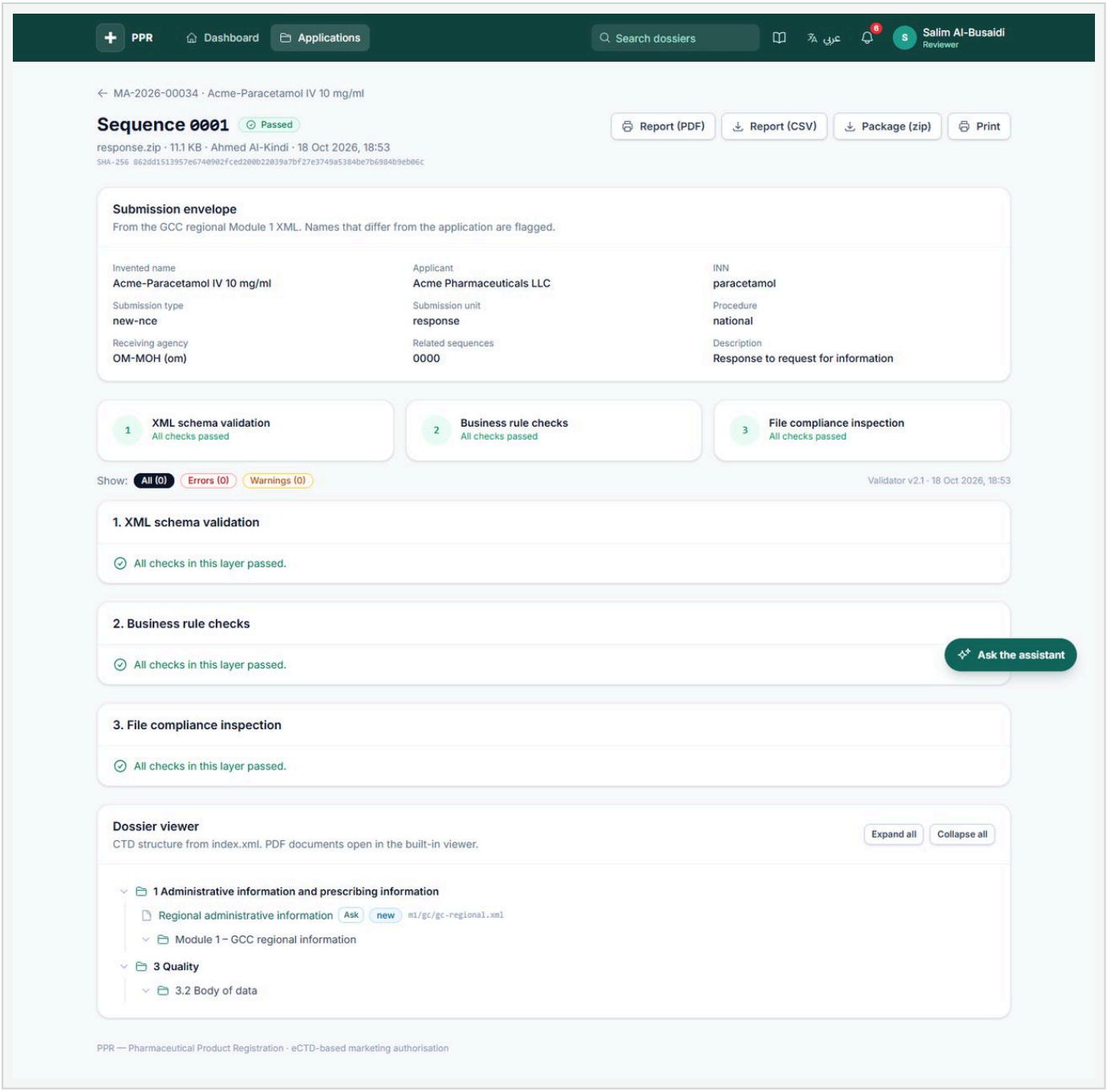


Figure 8: Passed report for response sequence 0001, with the GCC envelope compared to the application data.

6.3 Review and decision

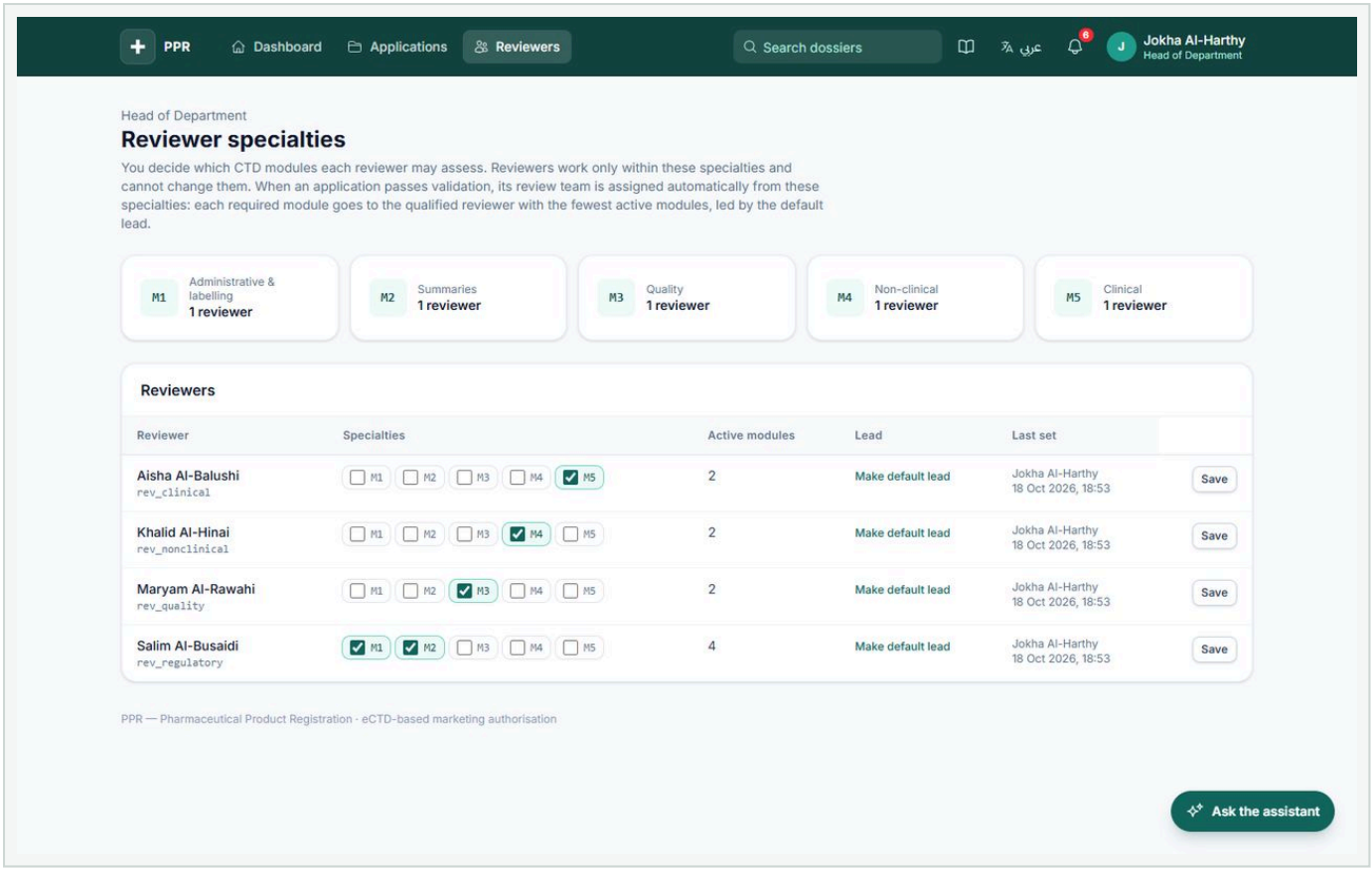


Figure 9: Reviewer specialties set by the Head of Department, with module coverage and the default lead.

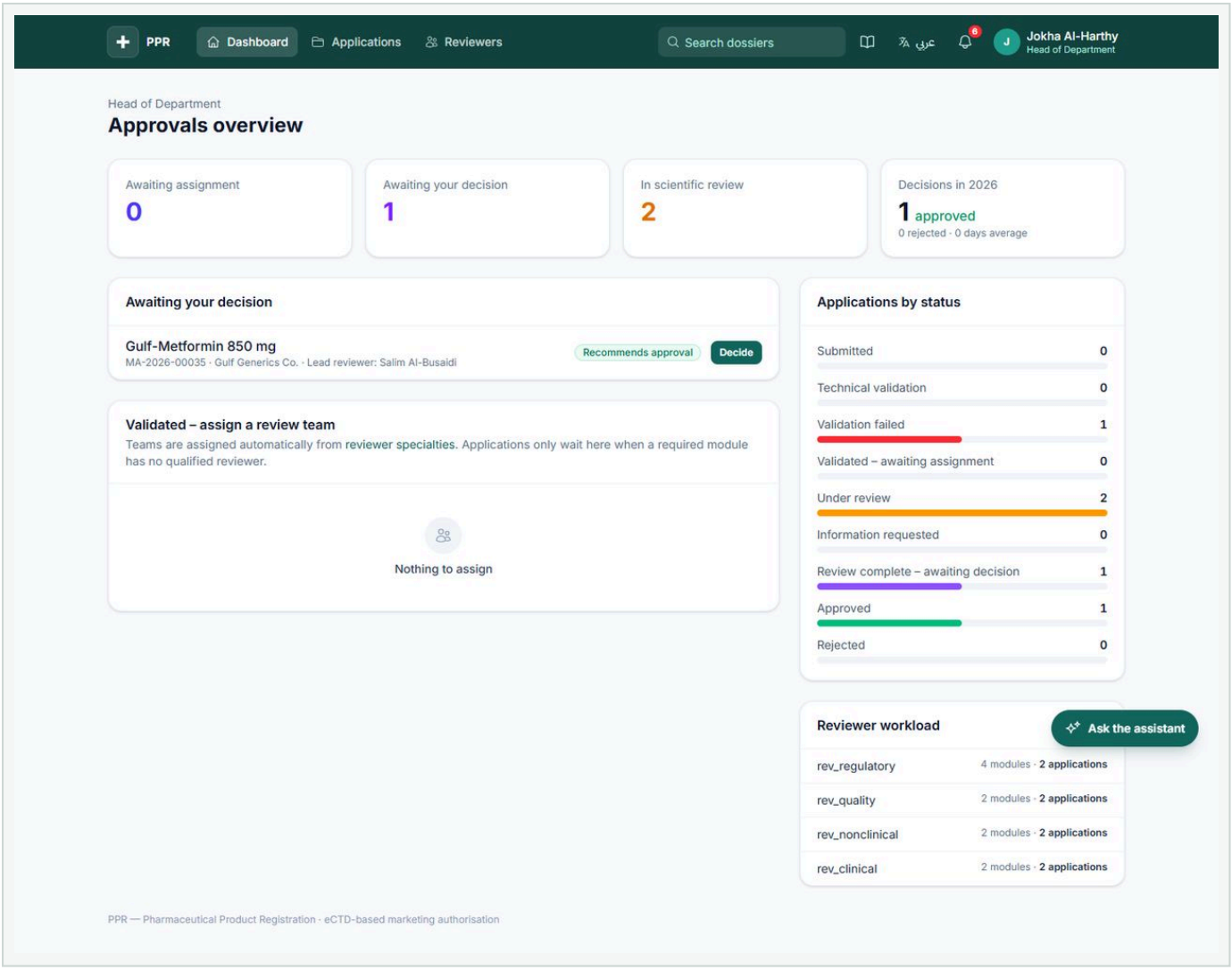


Figure 10: Head of Department dashboard: pending decisions, status summary and reviewer workload.

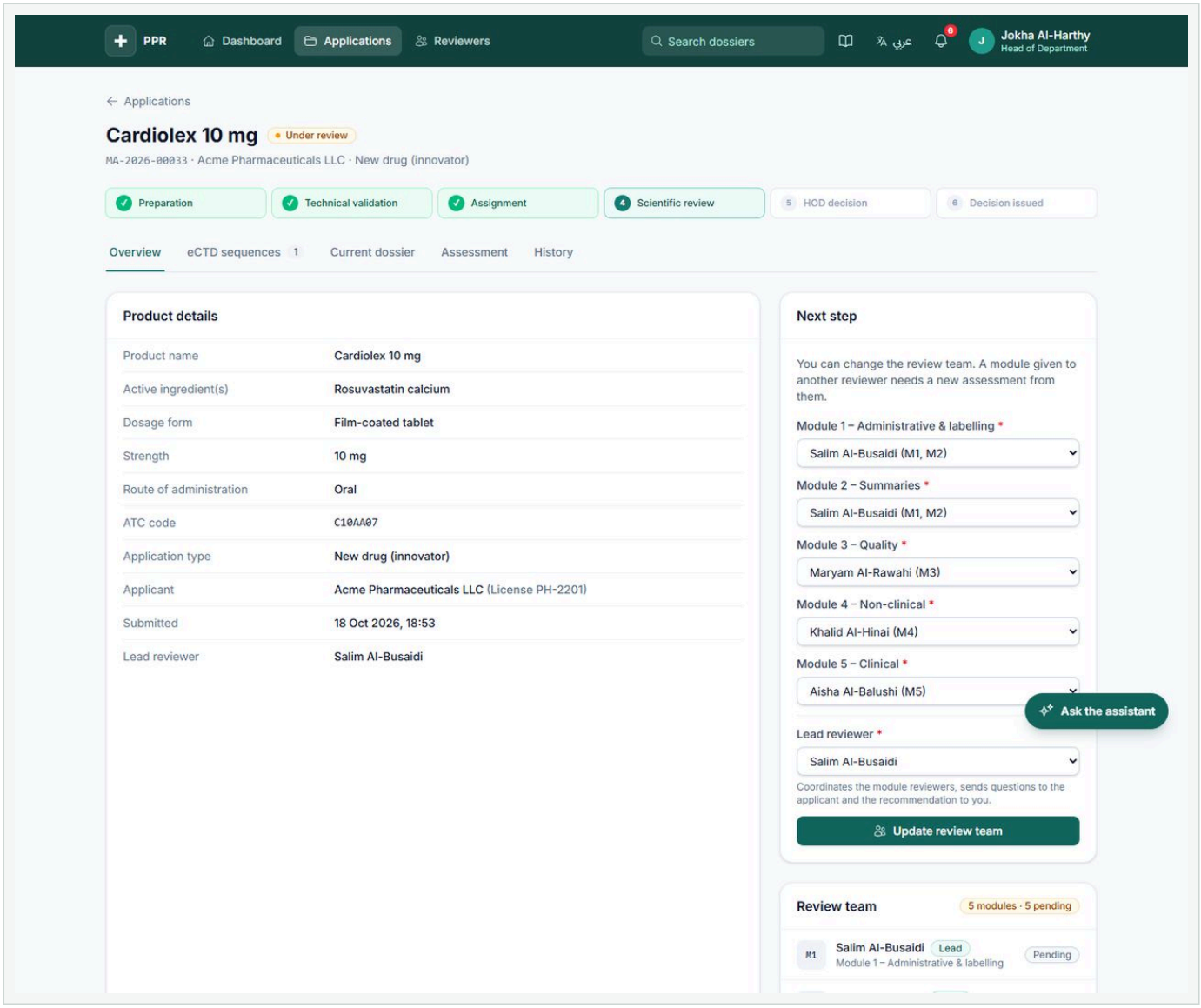


Figure 11: Application overview with progress indicator and the review team: one specialist per CTD module and a lead reviewer.

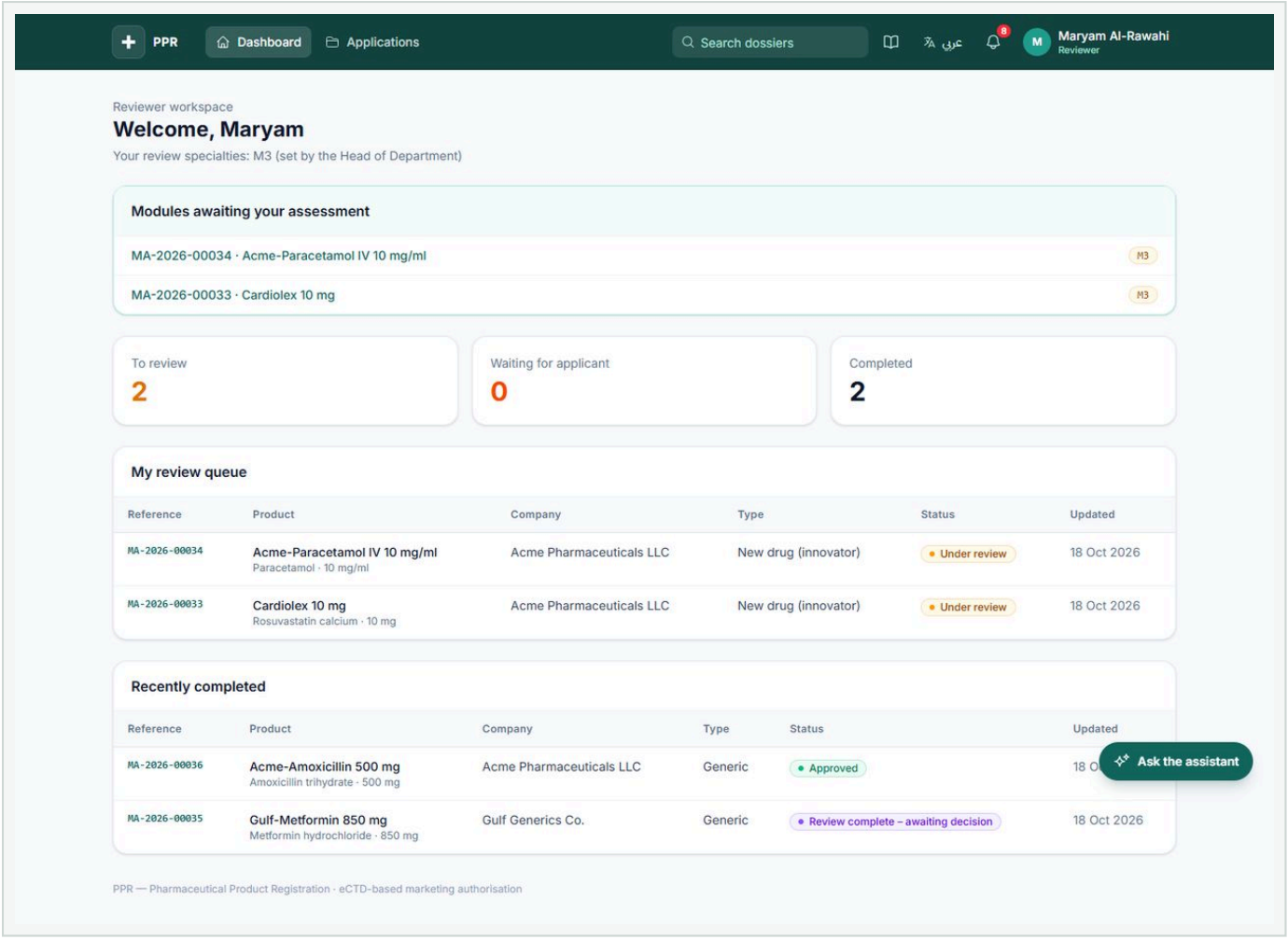


Figure 12: Reviewer dashboard listing the modules awaiting the reviewer's assessment.

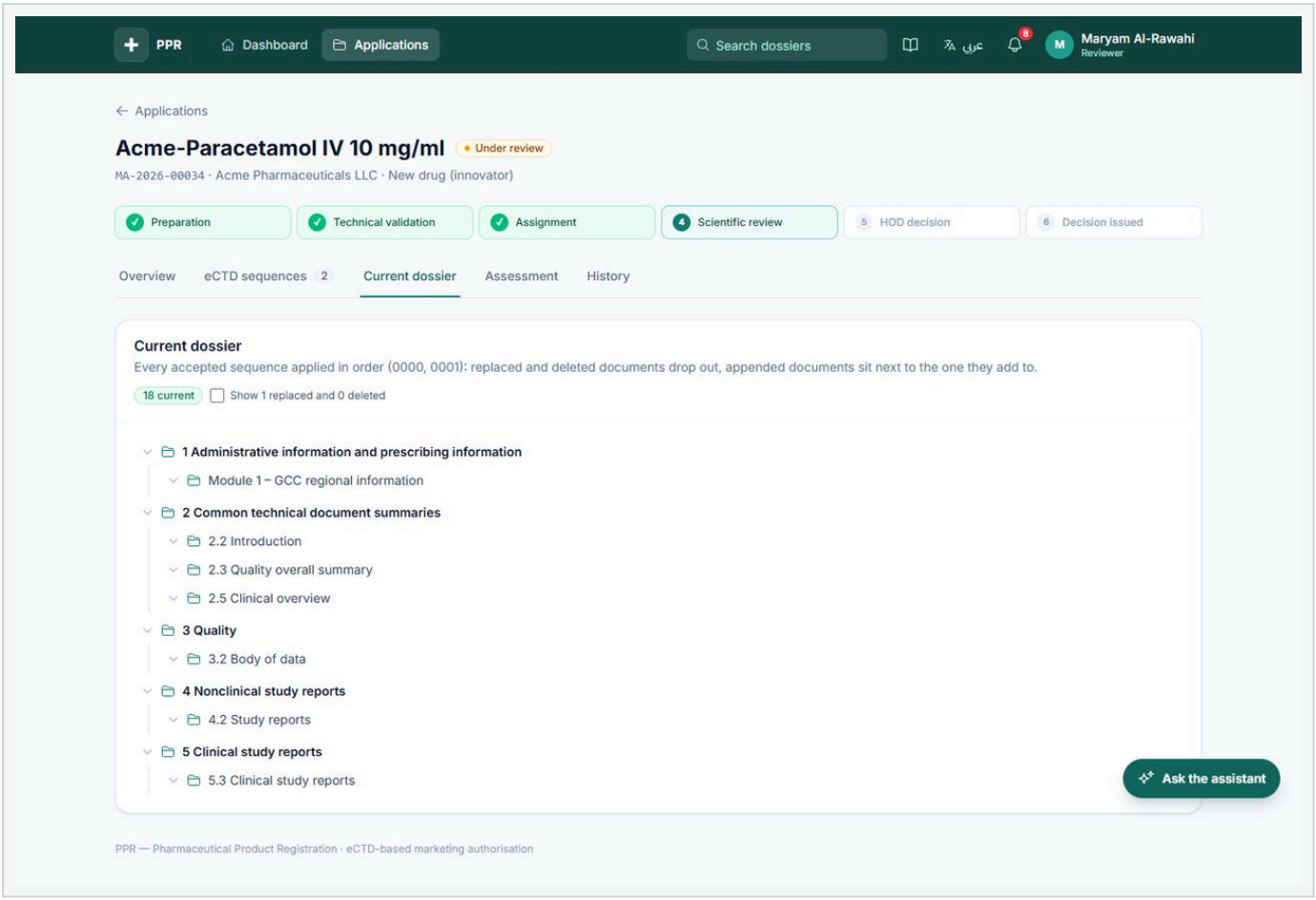


Figure 13: Current dossier: every accepted sequence applied in order, with replaced documents marked.

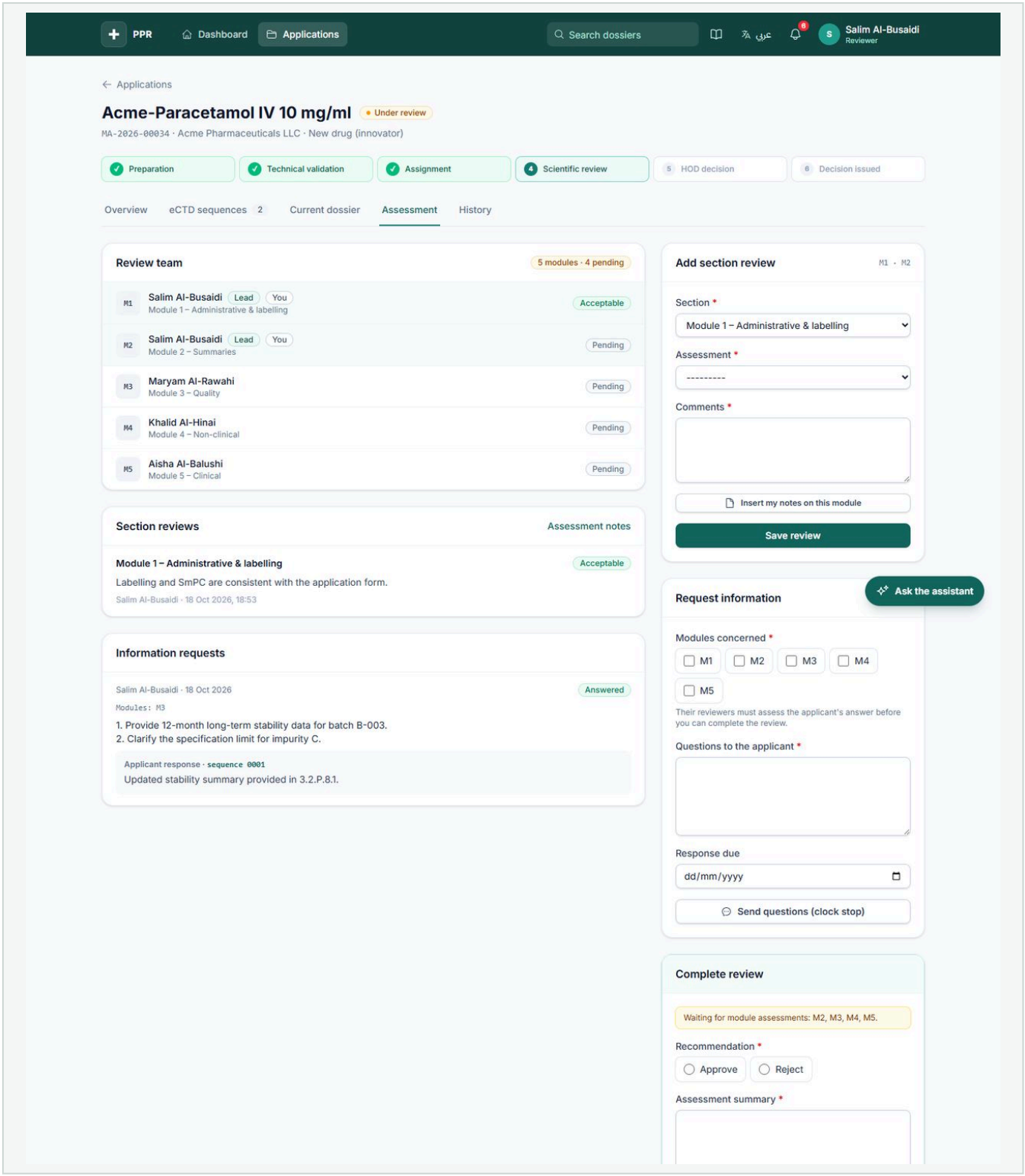


Figure 14: Assessment tab for the lead reviewer: review team progress, module reviews, information request with the applicant's response, and the recommendation form.

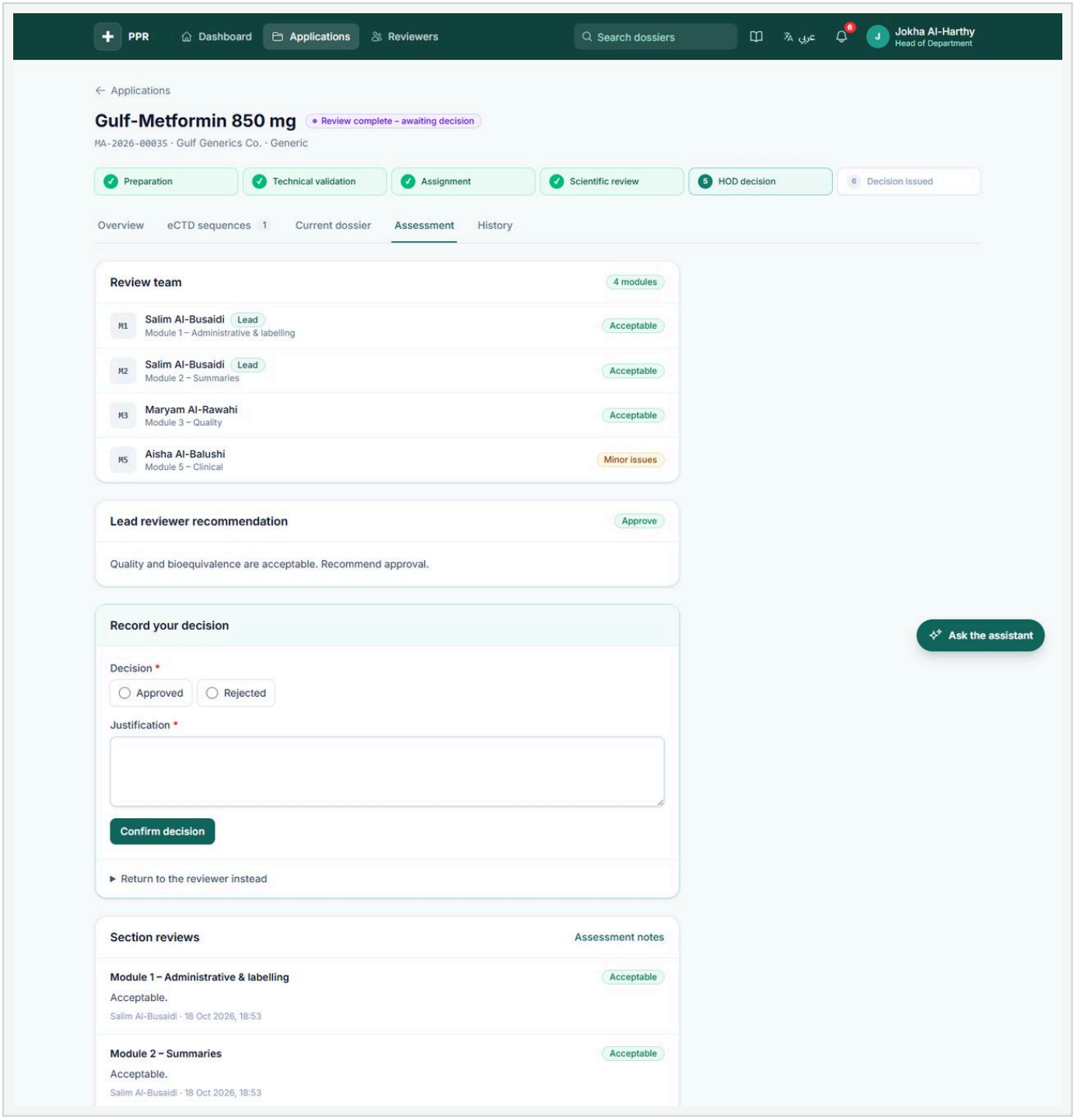


Figure 15: Head of Department decision based on the lead reviewer's recommendation and the module reviews.

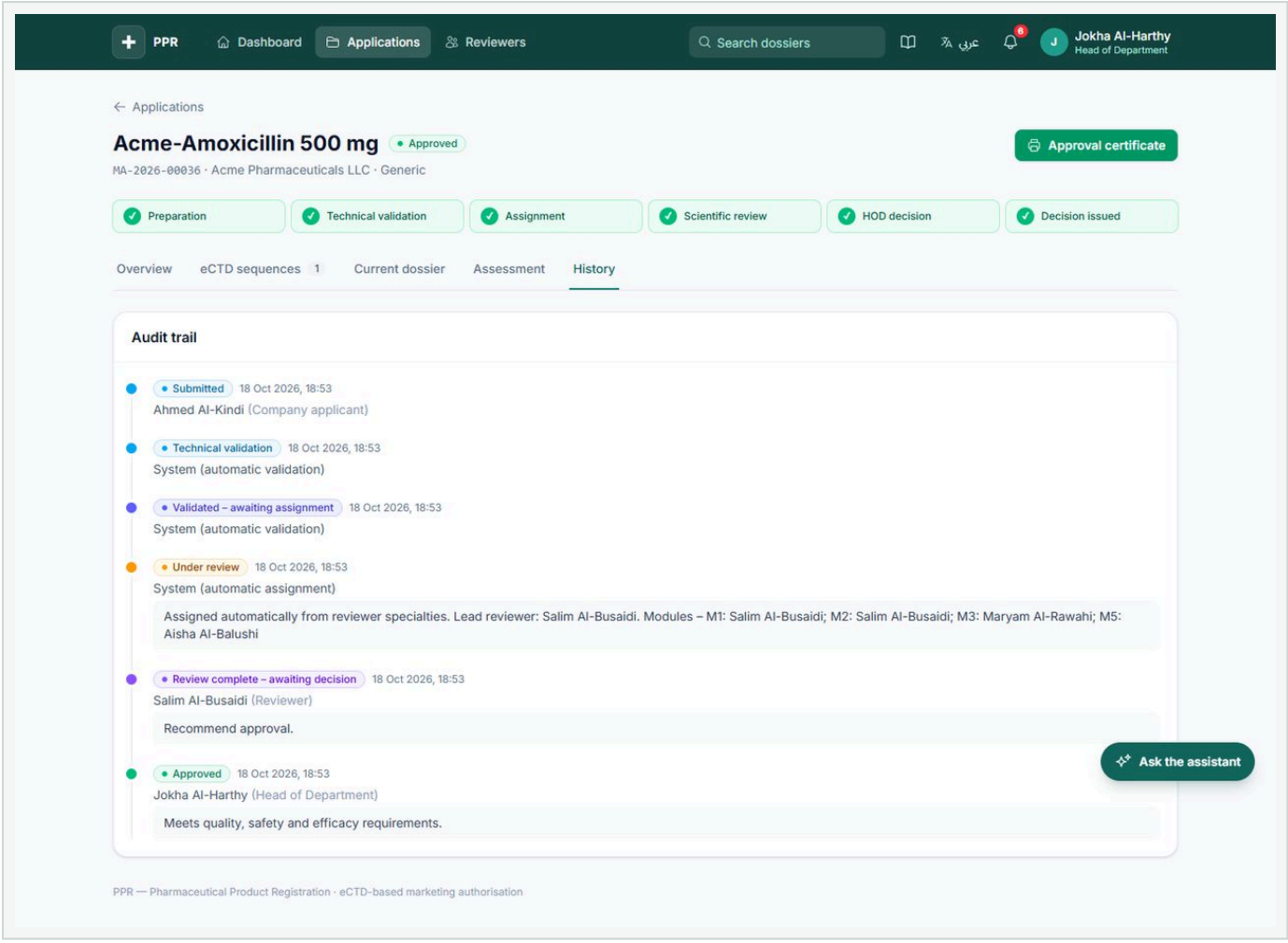


Figure 16: Audit trail of every status change, with user, time and note.

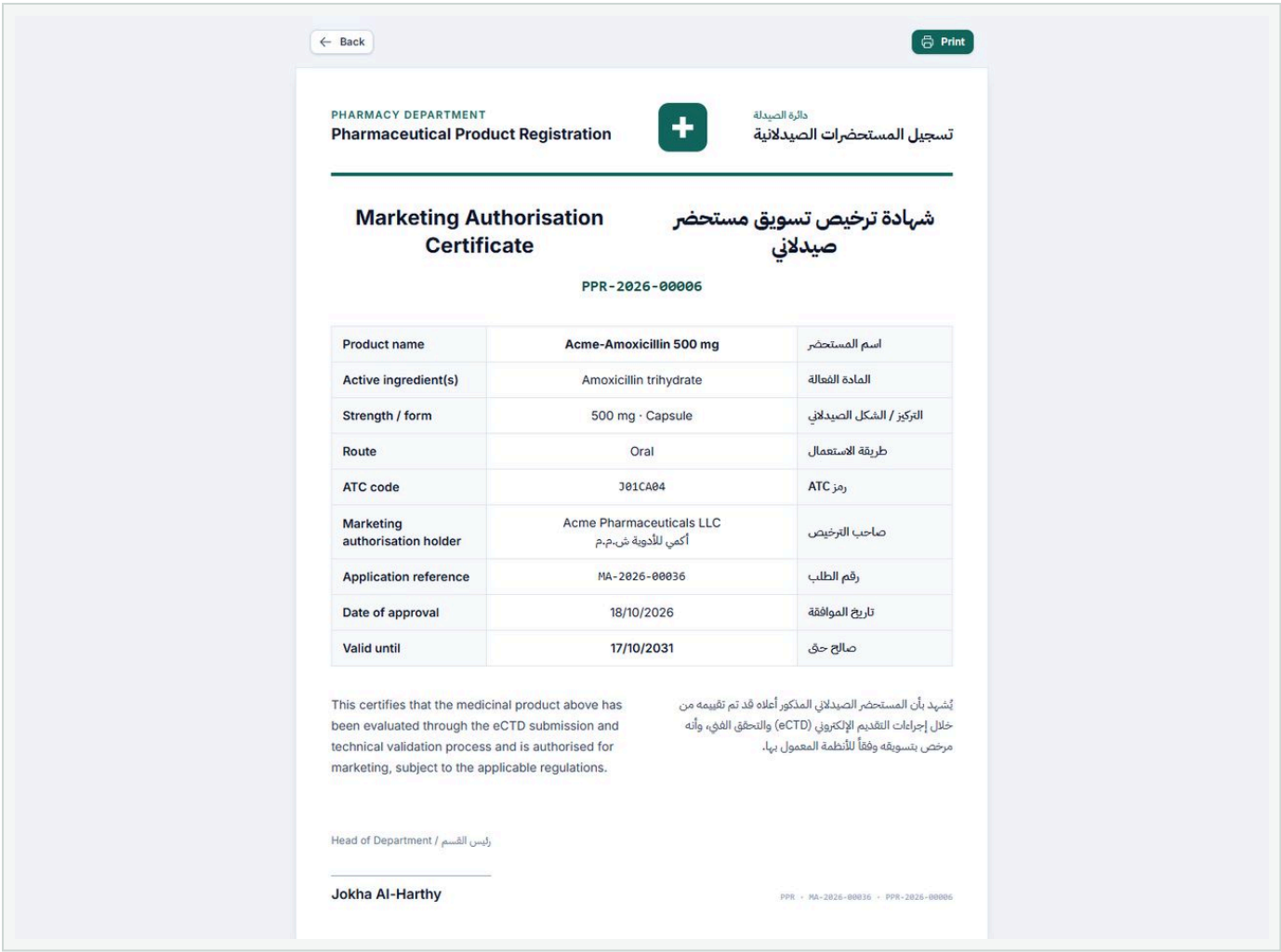


Figure 17: Bilingual marketing authorisation certificate.

6.4 Arabic interface and mobile layout

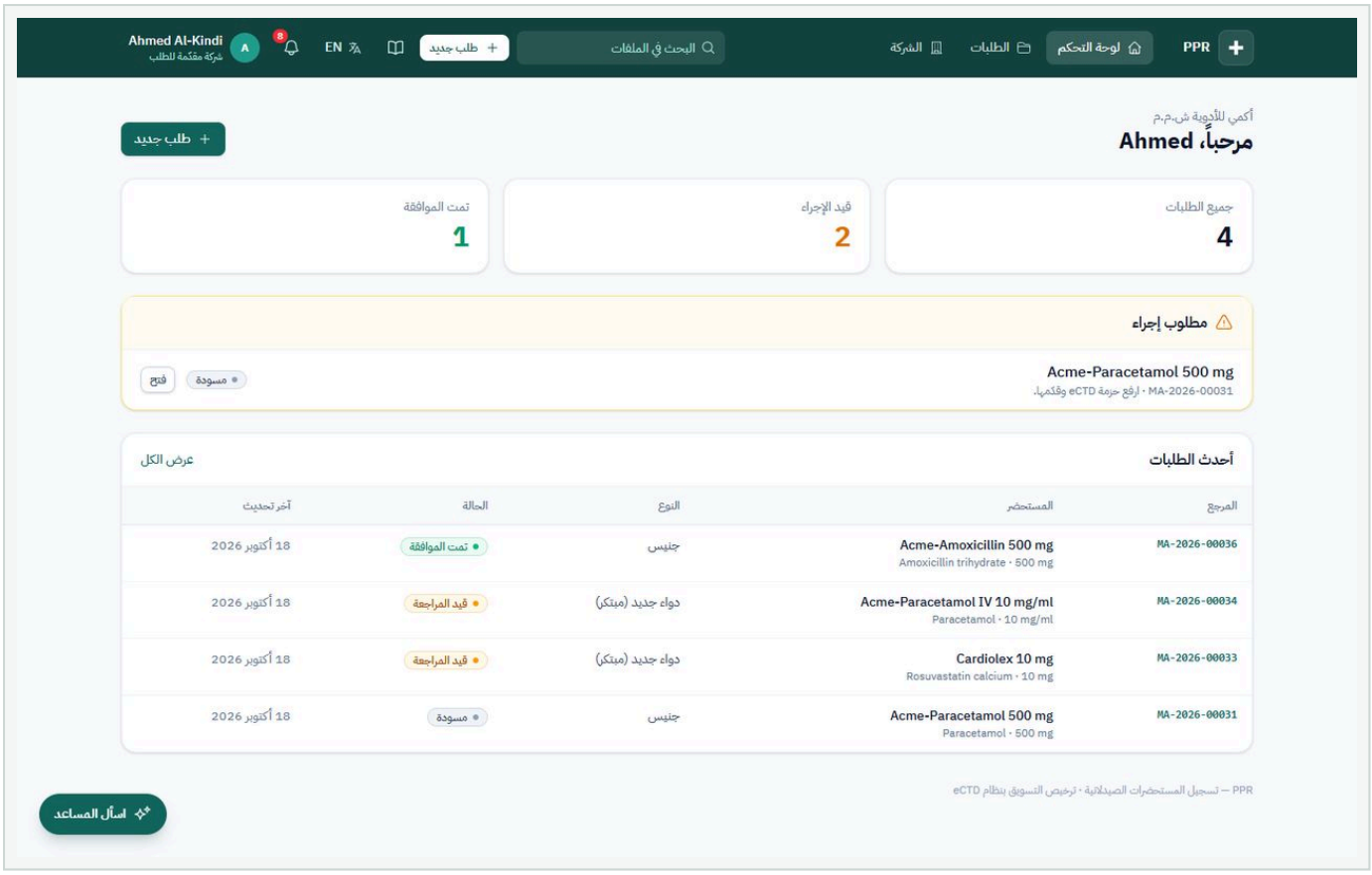


Figure 18: Company dashboard in Arabic with a right-to-left layout.

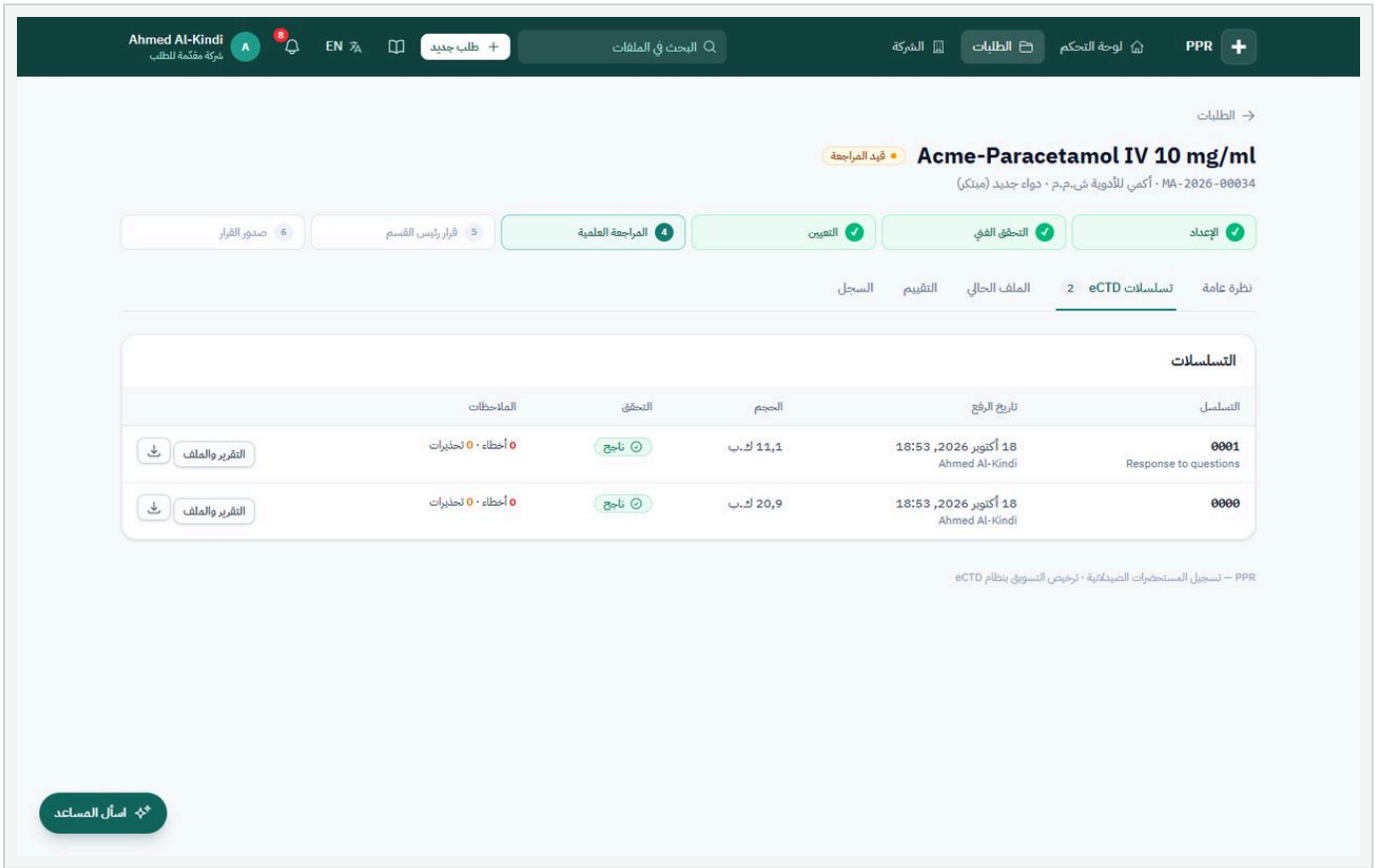


Figure 19: eCTD sequences page in Arabic.

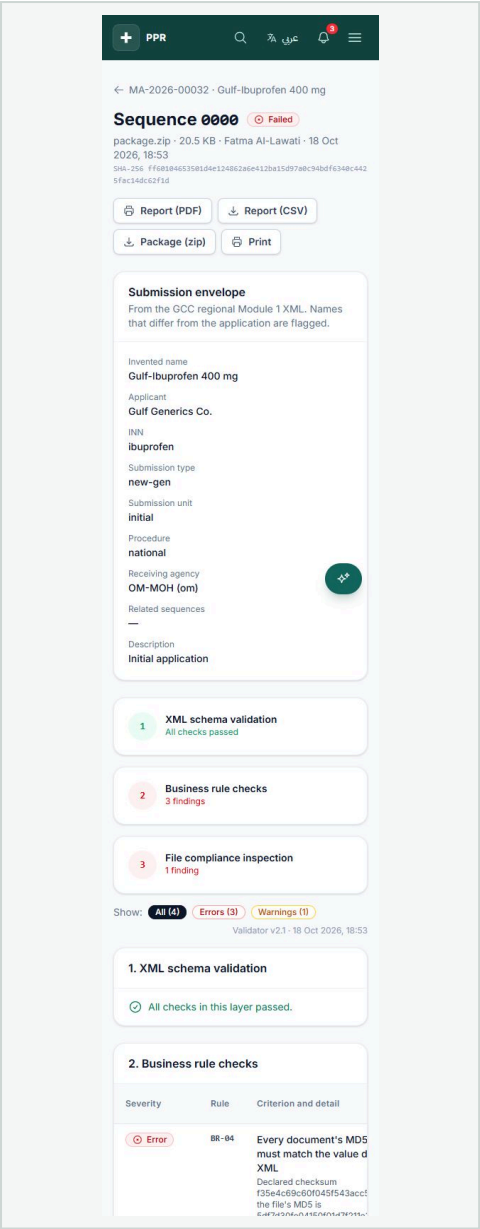


Figure 20: Validation report on a phone.

All pages adapt to phone screens without horizontal scrolling; wide tables scroll within their own area.

7 Workflow training guide

PPR includes a training guide at `/guide/`, opened from the book icon in the top bar (or the account and mobile menus). It explains every workflow in the system step by step: which role performs each step, the exact button or screen to use, the status the application moves to, the expected result, and practical tips. Button and screen names in the guide match the interface word for word, in English and in Arabic.

7.1 Contents of the guide

- **How an application moves**: the six stages of the progress bar, the statuses in each and who moves the application on, plus the three ways it goes back a step.
- **Twelve workflows** (Table 3), each with numbered steps labelled by role, the status change, the result and tips.
- **Status reference**: what each of the ten statuses means and who acts next.
- **Who does what**: a matrix of workflows against roles.

7.2 Using it for training

- Filter the guide by role, so each audience reads only its own workflows.
- Print it or save it as PDF from the page for classroom sessions.
- Practise on the demonstration accounts created by `seed_demo`: one company per applicant workflow, one reviewer per discipline and the Head of Department.
- The guide is generated from the same status names as the system and is covered by automated tests, so it stays in step with the workflow.

No.	Workflow	Roles	Status change
1	Sign in and manage your account	All	—
2	Create an application and submit the eCTD package	Company	Draft → Technical validation
3	Technical validation and correcting a failed package	Company, System	Technical validation → Validated
4	Set reviewer specialties	HOD	—
5	Assign the review team	HOD, System	Validated → Under review
6	Assess your CTD modules	Reviewer	Under review
7	Request information from the company (clock stop)	Lead reviewer	Under review → Information requested
8	Respond to an information request	Company, System	Information requested → Under review
9	Complete the review and send the recommendation	Lead reviewer	Under review → Review complete
10	Decide and issue the certificate	HOD, Company	Review complete → Approved / Rejected
11	Follow an application: notifications, history and search	All	—
12	Use the dossier assistant and review notes	All	—

Table 3: Workflows covered by the training guide.

Audience	Suggested session	Workflows
Company applicants	Prepare, submit and correct a dossier; answer questions; print the certificate	`1, 2, 3, 8, 10, 11`
Reviewers	Assess modules in the dossier viewer; as lead, send questions and the recommendation	`1, 6, 7, 9, 11, 12`
Head of Department	Specialties, review teams, decisions and monitoring	`1, 4, 5, 10, 11, 12`

Table 4: Suggested training sessions by audience.

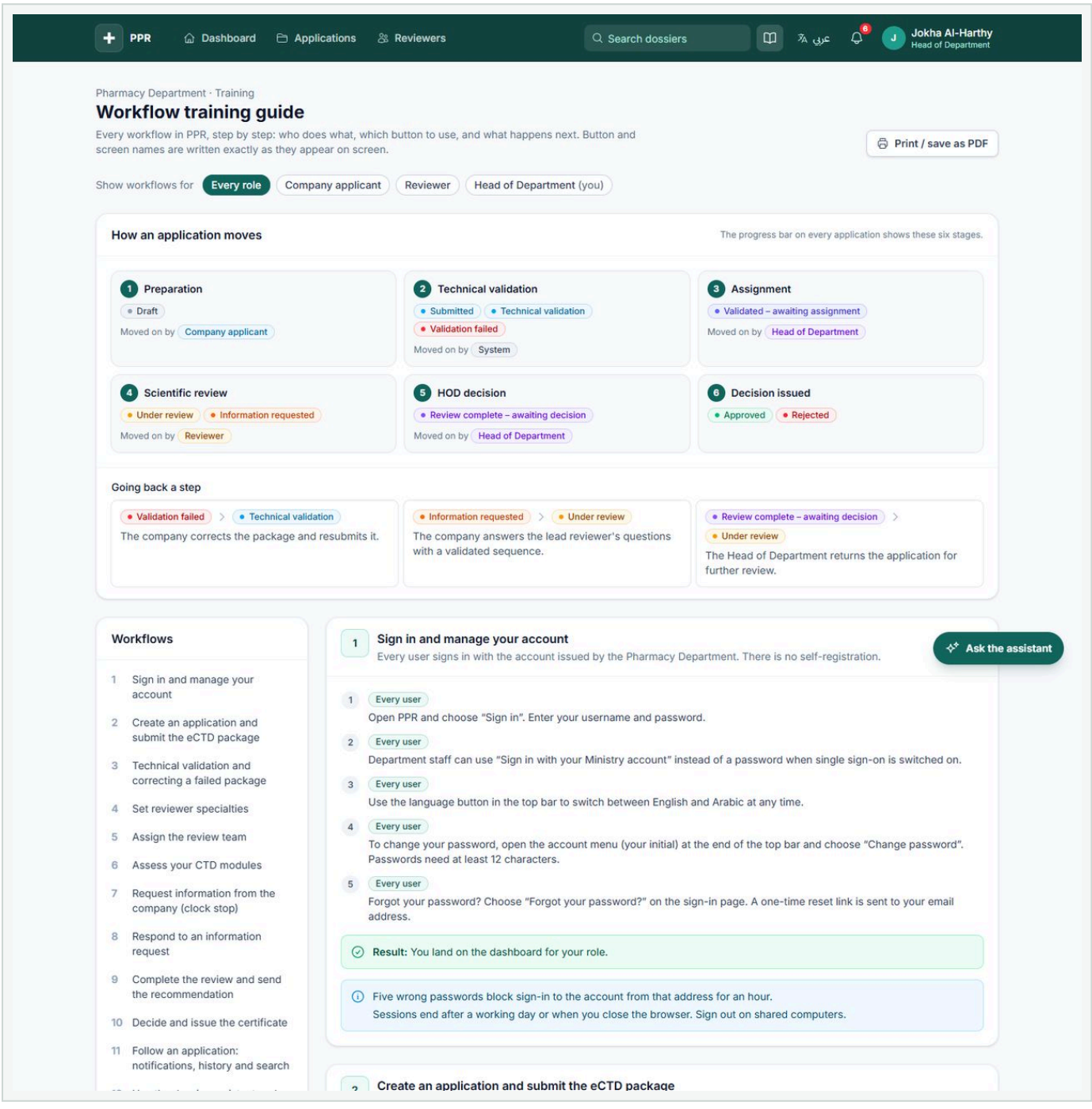


Figure 21: Workflow training guide: role filter, the six stages and the ways back a step.

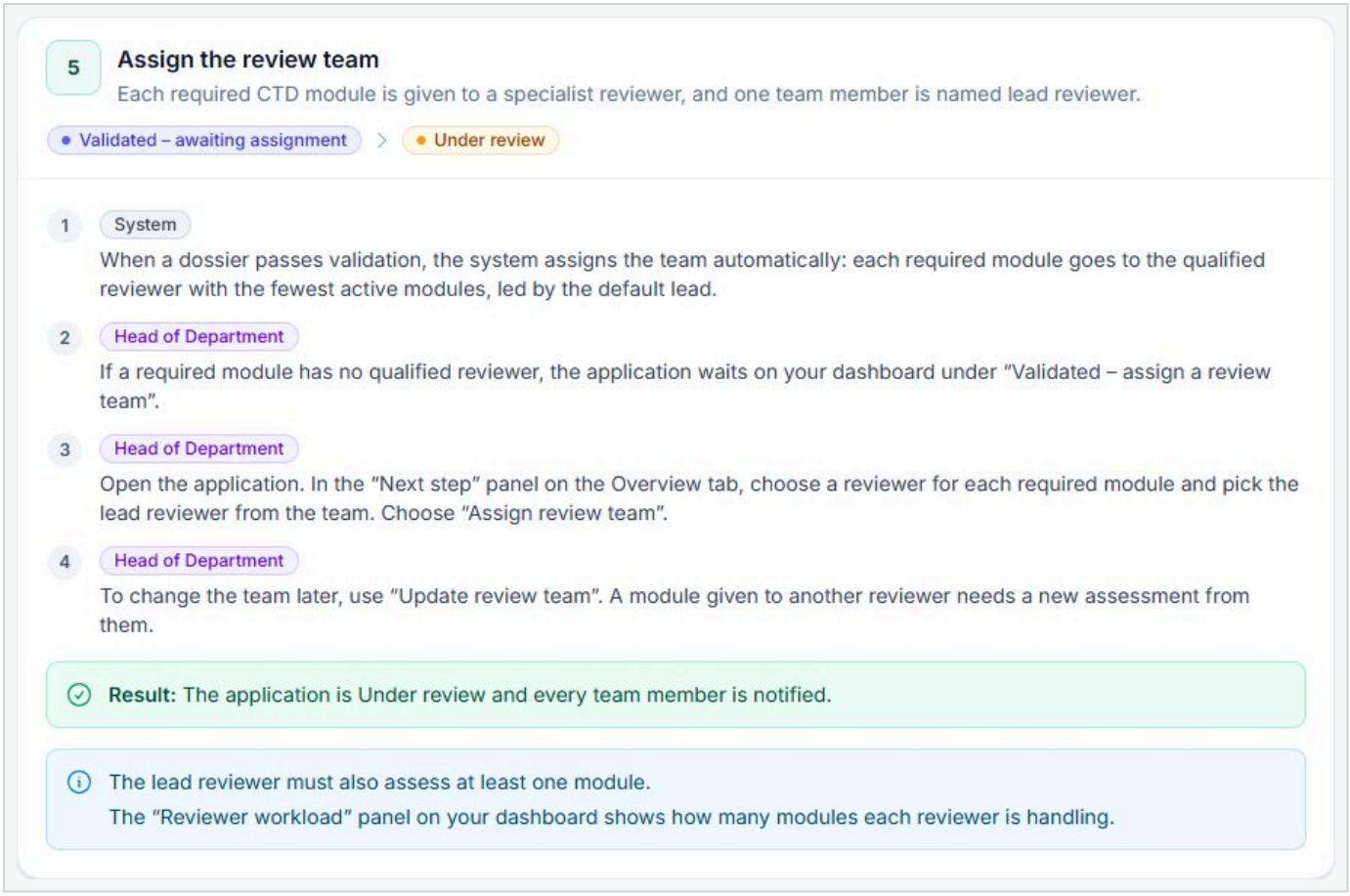
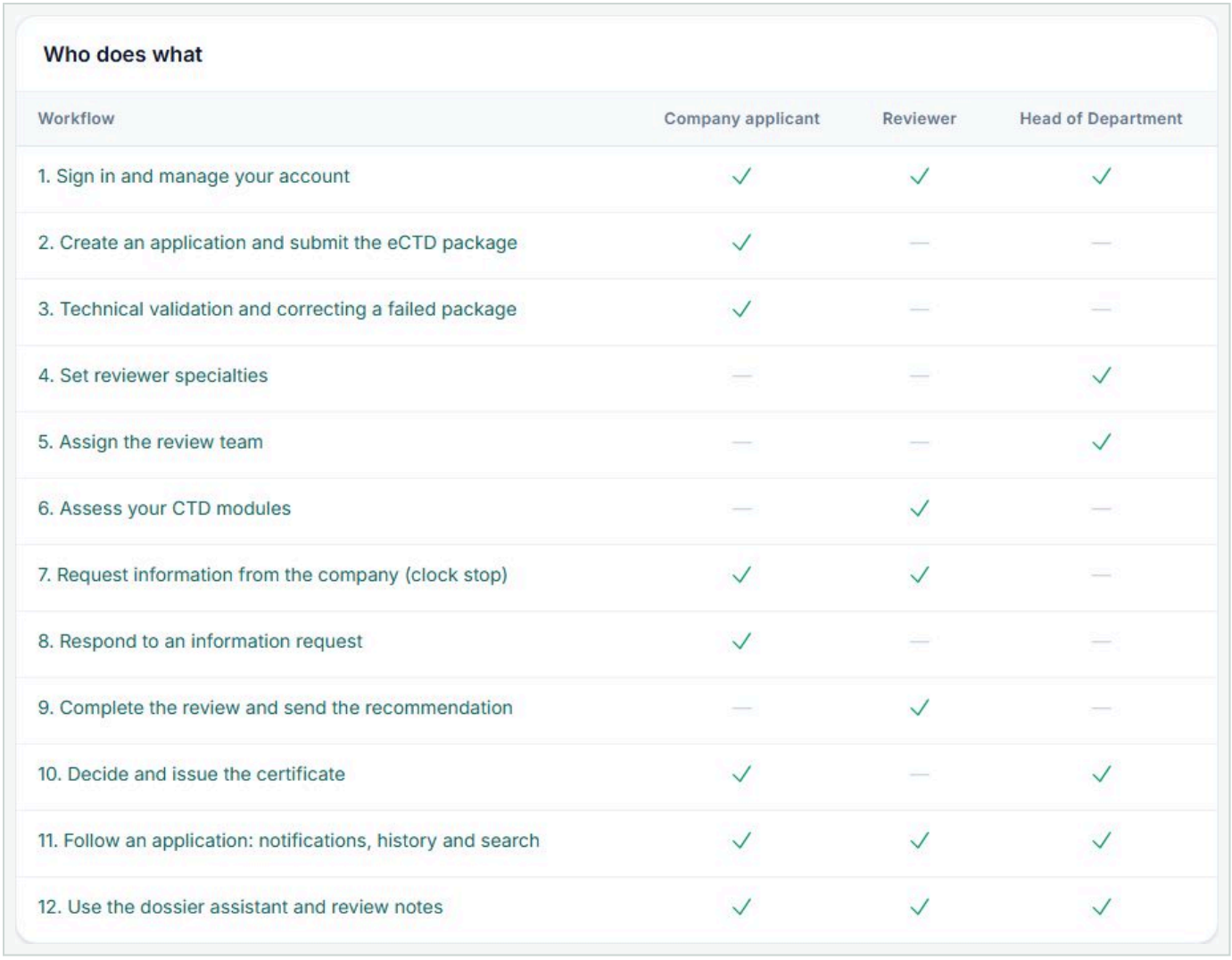


Figure 22: One workflow in the guide: steps by role, the status change, the result and tips.



Who does what

Workflow	Company applicant	Reviewer	Head of Department
1. Sign in and manage your account	✓	✓	✓
2. Create an application and submit the eCTD package	✓	—	—
3. Technical validation and correcting a failed package	✓	—	—
4. Set reviewer specialties	—	—	✓
5. Assign the review team	—	—	✓
6. Assess your CTD modules	—	✓	—
7. Request information from the company (clock stop)	✓	✓	—
8. Respond to an information request	✓	—	—
9. Complete the review and send the recommendation	—	✓	—
10. Decide and issue the certificate	✓	—	✓
11. Follow an application: notifications, history and search	✓	✓	✓
12. Use the dossier assistant and review notes	✓	✓	✓

Figure 23: Who does what: workflows against roles.

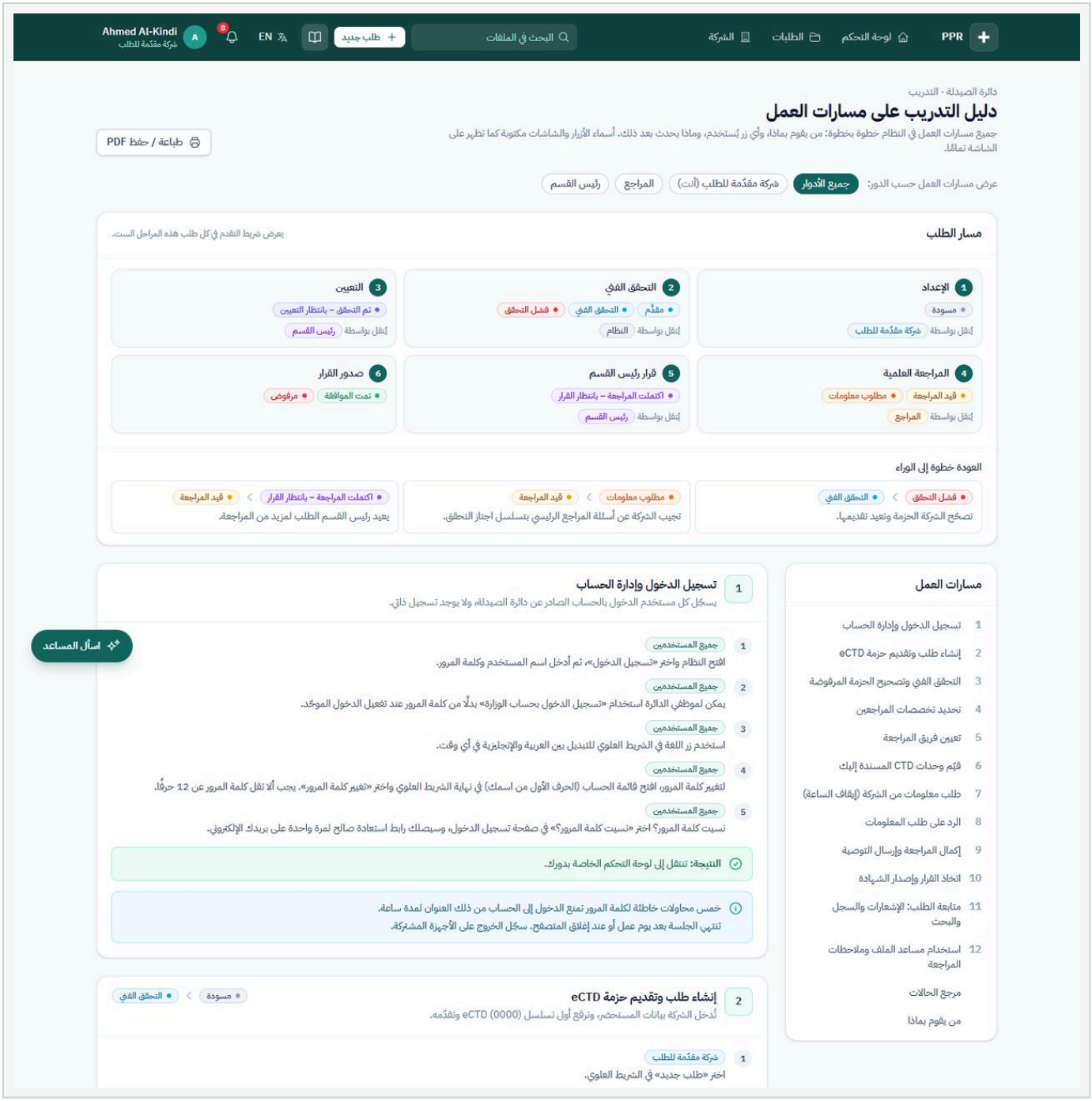


Figure 24: The training guide in Arabic.

8 Technical architecture

PPR is a single Django web application. In production, validation runs in a background worker (`run_validation_worker`, a PythonAnywhere always-on task) that also extracts document text for search and the assistant; locally it runs within the submission request. Uploaded dossiers are never published under a public address; every download passes through an access check.

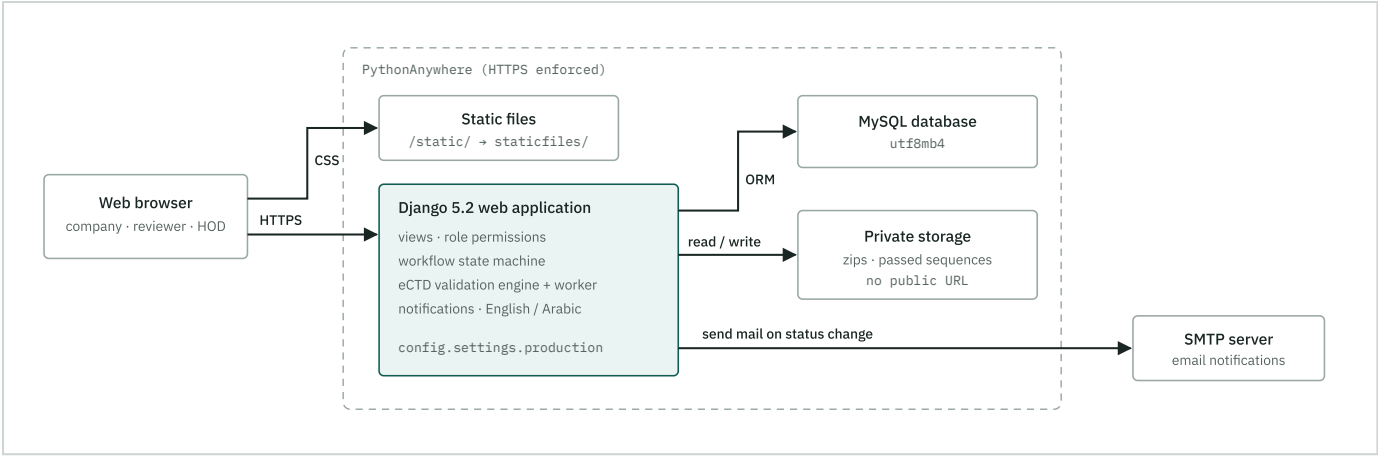


Figure 25: Deployment architecture on PythonAnywhere.

Component	Location in source code	Responsibility
Workflow	`apps/applications/workflow.py`	Allowed transitions per role, audit trail, notifications
Business services	`apps/applications/services.py`	Submit, respond, assign, review and decide
Validation engine	`apps/validation/`	Safe extraction, parsing, 29 rules (GCC Module 1 in `regions/`), findings, dossier tree, background worker
Review team and lifecycle	`apps/applications/`	Automatic team assignment, cumulative dossier view, review notes, search
Dossier assistant	`apps/assistant/`	Read-only tools limited to what the reader may open; answers cite document and page
API	`apps/api/`	Key-protected, read-only JSON for other systems
Training guide	`apps/core/guide.py`	Workflow training content shown at `/guide/`
Accounts	`apps/accounts/`	Users with role, company and review specialties; access control, lockout, optional single sign-on
Notifications	`apps/notifications/`	In-app notifications and email in the user's language
User interface	`templates/, theme/`	Tailwind templates supporting left-to-right and right-to-left

Table 5: Main software components.

9 Technology stack

Area	Technology	Version	Reason for selection
Programming language	Python	`3.12`	Supported by PythonAnywhere; mature XML and PDF libraries
Web framework	Django	`5.2 LTS`	Long-term support until 2028; built-in admin, authentication, translation and security
Styling	Tailwind CSS (django-tailwind)	`4.3 / 4.6`	Standalone build without Node.js
Interactivity	HTMX, Alpine.js	`2.0.4 / 3.14.8`	Lightweight interactions without a separate front-end application; served from the site
PDF viewer	PDF.js	`4.10`	In-app document viewer with text selection for review notes
Sign-in security	django-axes, mozilla-django-oidc	`8.3 / 4.0`	Lockout after repeated failures; optional single sign-on through OpenID Connect
XML processing	lxml	`6.1`	DTD validation with a parser hardened against external entities
PDF inspection	pypdf	`6.20`	Checks encryption, version, scripts and fonts
Database	SQLite (development), MySQL (production)	`mysqlclient 2.2`	No setup locally; MySQL is provided by PythonAnywhere
Configuration	django-environ	`0.14`	Secrets and database settings kept outside the code
Localisation	Django i18n with Babel and polib tooling	`EN / AR`	777 interface strings translated into Arabic
Typefaces	Inter, IBM Plex Sans Arabic	`Google Fonts`	Consistent Latin and Arabic text

Area	Technology	Version	Reason for selection
Testing	pytest, pytest-django	`9.1 / 4.14`	Automated tests on an in-memory database
Hosting	PythonAnywhere	`WSGI`	Managed Python hosting with HTTPS and MySQL

Table 6: Technology stack of the proof of concept.

9.1 Data model

- **Company**: registration and licence numbers, English and Arabic names
- **User**: role, company, review specialties, default lead
- **Application**: product details, type, status, lead reviewer
- **Review assignment**: one specialist per CTD module
- **Sequence**: one eCTD submission with SHA-256 and validation status
- **Validation job, run and finding**: the queue and the result of every rule
- **Document note and document text**: review notes and the search index
- **Review, information request, decision**: assessment and outcome
- **Status history, notification**: audit trail and alerts

9.2 Operational tools

- `seed_demo`: demonstration users and applications at each stage
- `build_sample_packages`: valid and invalid eCTD test packages
- `validate_sequence`: re-runs validation, for example after a DTD update
- `run_validation_worker`: background validation and text indexing
- `create_api_key`, `axes_reset`: API keys and sign-in lockouts
- `benchmark_validation`, `scripts/traceability.py`: validation evidence
- `scripts/i18n.py`: extracts and compiles translations
- `docs/deploy_pythonanywhere.md`: deployment procedure

10 Security and quality assurance

10.1 Security controls

- Role checks on every page: companies see only their own applications, reviewers only the applications of their review teams, and assess only their own modules.
- Status changes only through the state machine, each recorded in the audit trail.
- Package extraction blocks path traversal, symbolic links, encrypted archives and oversized archives.
- XML read with external entities, DTD loading and network access disabled.
- Dossier documents served only after an access check, with protective security headers.
- Production settings enforce HTTPS with HSTS, secure cookies, CSRF and clickjacking protection, and a nonce-based Content-Security-Policy; Django's deployment check passes.
- Sign-in lockout after five failures, passwords of at least 12 characters, reset by email, 8-hour sessions; optional single sign-on for staff.
- Review notes are internal to the department and never shown to applicants.

10.2 Testing

Test area	Result
Valid packages pass all rules; one invalid package per rule fails it	Passed
Full lifecycle, resubmission, response and return paths	Passed
Review teams: specialties, automatic assignment, lead-only actions	Passed
Permissions per role and confidential review notes	Passed
All pages per role, English and Arabic, including the training guide	Passed
Security: lockout, password rules, protected downloads, API keys, SSO	Passed
Background validation queue, search and assistant tools	Passed
Total: 149 automated tests	**149 passed**

Table 7: Automated test results.

11 Limitations and recommendations

Required before accepting real submissions: the official ICH `ich-ectd-3-2.dtd` is now bundled unchanged. Still open: replace the GCC regional DTD, transcribed from the GCC Module 1 specification, with the files distributed by SFDA; confirm the list of required Module 1 documents

(rule GCC-07) with the Drug Safety Center; and decide where the dossier assistant's language model may run before enabling it with real dossiers.

No.	Recommendation	Phase
R1	Adopt the official ICH DTD and regional Module 1 specification. *ICH DTD and GCC M1 v1.5 rules done; distributed GCC DTD files pending.*	Pilot
R2	Run validation in the background for large dossiers. *Done (validation worker).*	Pilot
R3	Company self-registration with document verification	Pilot
R4	Single sign-on or two-factor login for staff. *Single sign-on done; two-factor pending.*	Pilot
R5	Backup, retention and storage policy for dossiers	Pilot
R6	Combined lifecycle view of all sequences in a dossier. *Done (Current dossier tab).*	Phase 2
R7	Review timelines with target days and reports	Phase 2
R8	QR code on certificates for public verification	Phase 2
R9	Fees and payments; variations and renewals linked to approvals	Phase 2
R10	Accessibility review and load testing	Phase 2
R11	Run role-based training sessions with the workflow training guide before the pilot	Pilot

Table 8: Recommendations for the next phases.

12 Approval

PREPARED BY

Name

Signature

Date

REVIEWED BY

Name

Signature

Date

APPROVED BY · HEAD OF DEPARTMENT

Name

Signature

Date

A Glossary

Term	Meaning
CTD	Common Technical Document: the ICH structure of five modules (administrative, summaries, quality, non-clinical, clinical).
eCTD	The electronic form of the CTD, with an XML backbone listing every document.
Sequence	One eCTD submission, numbered 0000 for the initial dossier and 0001 onwards for later submissions.
DTD	Document Type Definition: the rules an eCTD `index.xml` must follow.
MD5 checksum	A fingerprint of a file used to confirm it has not changed.
Lifecycle operation	How a document relates to earlier sequences: new, replace, append or delete.
ICH	International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use.
GCC Module 1	The regional Module 1 (administrative information) used by the Gulf states, version 1.5.
Lead reviewer	The member of the review team who sends questions and the team's recommendation.
Clock stop	The pause in the review while the applicant answers questions.
HOD	Head of Department.